MTR & COA Automation: A Buyer's Guide for Steel & Metals Companies

How to automate certificate extraction, validation, traceability and ERP workflows without compromising quality.

A practical guide for Quality, QA/QC, Operations, IT and Digital Transformation leaders

Contents

- Executive Summary
- 1. Why MTR & COA Processing Is Harder Than It Looks
- 2. The Reality of Supplier Documents
- 3. What the Automation System Must Understand
- 4. Standards Complexity: ASTM, ASME, EN and Customer Rules
- 5. OCR Is Not the Same as Intelligent Automation
- 6. Certificate Validation: Where Automation Creates Real Quality Value
- 7. Traceability: Connect the Certificate to the Material
- 8. ERP, QMS and LIMS Integration
- 9. Exception Management & Human-in-the-Loop
- 10. Security, Governance & Compliance
- 11. Building the Business Case
- 12. Build vs. Buy
- 13. Vendor Evaluation Framework
- 14. 20-Point MTR/COA Automation Vendor Evaluation Checklist
- 15. Pilot / PoC Checklist
- 16. What a Future-Ready MTR/COA Process Looks Like
- About Star Software
- Call to Action

Executive Summary

MTRs (Mill Test Reports) and COAs (Certificates of Analysis) are among the most operationally important documents in the steel and metals supply chain. They contain the evidence needed to verify material identity, chemistry, mechanical properties, specifications and traceability—but they often arrive as PDFs, scans, spreadsheets and supplier-specific templates.

The challenge is not simply reading these documents. A useful automation system must identify the document, extract the right fields, understand context, validate values against applicable rules, preserve traceability, route exceptions to people and deliver trusted data into the systems where quality and operational decisions are made.

This buyer's guide provides a practical framework for evaluating MTR/COA automation. It focuses on the capabilities that matter to steel and metals organizations: document variability, heat numbers and grades, ASTM/ASME/EN requirements, chemical and mechanical properties, validation, traceability, ERP/QMS/LIMS integration, exception handling, human review, security and governance.

Core principle: Extraction is only step one. Trusted, validated data is the real outcome.

1. Why MTR & COA Processing Is Harder Than It Looks

On paper, the process appears straightforward: receive a certificate, read the relevant fields, check them and record the results. In practice, the document is a variable interface between suppliers and the buyer's quality system.

Typical sources of complexity include:

2. The Reality of Supplier Documents

- Different supplier templates and field labels
- Native PDFs mixed with scanned documents
- Multi-page certificates and supporting test reports
- Tables with merged cells, unusual layouts or varying column structures
- Different units, abbreviations and naming conventions
- Missing or optional fields
- Stamps, signatures, annotations or low-quality scans
- Certificates received by email, portals, shared folders or other channels

Supplier documents can differ substantially even when they describe the same material. A certificate may be digitally generated, scanned from paper, emailed as an attachment, embedded in a multi-page packet or accompanied by supporting test reports.

Common variations include:

3. What the Automation System Must Understand

- Heat number / heat identification
- Material grade and product designation
- Applicable specification or standard
- Product form, dimensions and size

- Chemical composition and element values
- Mechanical properties such as tensile strength, yield strength and elongation where applicable
- Test results and supplementary requirements
- Lot, batch, order, PO or shipment identifiers
- Supplier and mill information
- Relationships between the certificate and the material or transaction

A steel or metals certificate is not merely a collection of text fields. The relationship between identifiers, specifications, properties and test results matters.

4. Standards Complexity: ASTM, ASME, EN and Customer Rules

Standards create a validation context, not simply a label. A robust implementation should allow organizations to configure rules for applicable specifications, customer requirements and internal quality procedures.

Example validation matrix

Validation area	Example question	Typical source	Buyer expectation
Identity	Does the heat number match the receiving record?	ERP / certificate	Configurable cross-check
Grade	Is the reported grade consistent with the order?	PO / material master	Rules-based validation
Chemistry	Are reported elements within the applicable requirement?	Certificate / specification	Range-based checks
Mechanical	Are required properties present and within limits?	Certificate / specification	Configurable thresholds
Traceability	Can the certificate be linked to the material/lot?	ERP / receiving	Persistent linkage

5. OCR Is Not the Same as Intelligent Automation

OCR can turn pixels into text. That is useful, but it does not by itself determine whether a value is the right value, whether it belongs to the correct field, whether it satisfies a specification or what should happen when confidence is low.

Manual vs. OCR-only vs. intelligent document automation

Capability	Manual	OCR-only	Intelligent automation
Document reading	Human	Text capture	Classification + context
Field extraction	Human	Basic text/OCR	Structured extraction
Validation	Human	Usually limited	Configurable rules
Exceptions	Email/spreadsheet/manual	Often manual	Workflow + human review
Traceability	Manual linking	Partial	Record-level linkage
ERP/QMS/LIMS	Manual entry	Limited	Integration-ready output
Auditability	Depends on process	Limited	Workflow and change trail

6. Certificate Validation: Where Automation Creates Real Quality Value

- Field-level validation: Is the value present, correctly formatted and plausible?
- Rules-based validation: Does the value satisfy a configured business or quality rule?

- Specification validation: Does the reported result fall within the applicable requirement?
- Cross-field validation: Do related fields make sense together?
- Cross-document validation: Does the certificate agree with related records?
- Traceability validation: Can the certificate be linked to the relevant material, heat, lot or order?
- Confidence-based review: Which fields or documents require human attention?

Validation should operate at multiple levels. The objective is to identify exceptions before incorrect or incomplete information becomes part of downstream records.

7. Traceability: Connect the Certificate to the Material

Certificate automation becomes strategically valuable when certificate data can be connected to the physical material and the transaction that brought it into the organization.

Illustrative flow

Supplier document → Certificate classification → Heat/lot identification → Field extraction → Validation → Exception review (if required) → Material/order linkage → ERP/QMS/LIMS update → Audit trail

8. ERP, QMS and LIMS Integration

The automation layer should not become another isolated repository. Its purpose is to deliver trusted, structured information into the systems of record and workflows already used by the organization.

9. Exception Management & Human-in-the-Loop

The goal should not be to eliminate human involvement at any cost. The better objective is to reserve human attention for cases where judgment, ambiguity or risk genuinely requires it.

10. Security, Governance & Compliance

MTRs and COAs can contain commercially sensitive supplier, product and quality information. Buyers should evaluate security and governance as part of the automation architecture—not as an afterthought.

11. Building the Business Case

The business case should be built from the organization's own baseline. Avoid relying solely on generic claims about percentage savings.

Illustrative ROI framework

Annual processing cost = documents per year × average manual minutes per document × fully loaded labor cost per minute.

Potential automation value can then be assessed through reduced processing effort, lower rework, fewer data-entry errors, faster material release, improved audit readiness and greater throughput. Use your own baseline and measured pilot results rather than generic vendor percentages.

12. Build vs. Buy

Organizations should decide whether to build, configure or buy based on document complexity, internal expertise, integration requirements, governance expectations and the cost of maintaining the solution as suppliers and requirements change.

Question	Build	Buy / Configure
Do we have sustained AI/document engineering capacity?	Important	Less critical
How variable are supplier documents?	Higher ongoing maintenance	Vendor platform capability matters
How complex are ERP/QMS/LIMS workflows?	Requires internal integration ownership	Evaluate integration capability
How fast must value be demonstrated?	Often slower	Potentially faster with a focused pilot
Who owns model/rule maintenance?	Internal	Shared with vendor

13. Vendor Evaluation Framework

- Document variability: Can the platform handle the actual document population?
- Extraction quality: Can it reliably identify fields and tables?
- Validation: Can quality and business rules be configured?
- Workflow: Can exceptions be routed and resolved?
- Integration: Can trusted data reach ERP/QMS/LIMS and other workflows?
- Governance: Are access, audit, retention and change controls appropriate?
- Implementation: Can the vendor support a representative pilot and production rollout?

Evaluate vendors across five dimensions: document handling, extraction and AI, validation and quality, integration and workflow, and security/implementation.

14. 20-Point MTR/COA Automation Vendor Evaluation Checklist

#	Evaluation criterion	What good looks like	Questions to ask vendor	Score 1–5
1	Multi-format ingestion	Supports the document types and channels used by your suppliers.	Which formats and intake channels are supported?	
2	Layout variability	Handles different supplier templates without requiring a separate brittle workflow for every document.	How does the system handle new supplier layouts?	
3	Multi-page context	Maintains context across pages and related certificate documents.	Can fields be linked across pages and attachments?	
4	Field extraction	Extracts required certificate fields and tables with traceable source context.	Can we see the source location for an extracted value?	
5	Table extraction	Handles chemical and mechanical property tables reliably.	How are complex and variable tables processed?	
6	Confidence scoring	Provides usable confidence indicators to support review decisions.	How are low-confidence fields identified?	
7	Adaptability	Can the solution accommodate new suppliers, layouts and terminology.	What is the process and effort to onboard a new supplier?	
8	Configurable rules	Allows business and quality rules to be configured without rebuilding the entire solution.	Who configures validation rules and how are changes governed?	
9	Specification validation	Can compare certificate values with applicable specifications or customer requirements.	Can rules vary by material grade, specification or customer?	
10	Cross-field checks	Can detect inconsistencies between related fields.	Which cross-field validations are supported?	
11	Exception management	Routes exceptions to appropriate users with clear	How are exceptions presented, assigned and resolved?	

		reasons and evidence.		
12	ERP integration	Can deliver structured, validated data into the ERP or system of record.	Which integration patterns and APIs are available?	
13	QMS/LIMS integration	Can support quality workflows and laboratory/quality systems where required.	How does the platform integrate with QMS/LIMS?	
14	Traceability	Maintains links between certificate, heat/lot/material/order and downstream records.	How is end-to-end traceability preserved?	
15	Audit trail	Records extraction, validation, review and change events.	Can administrators retrieve a complete audit history?	
16	Access control	Supports appropriate role-based access and administrative controls.	How are roles, permissions and segregation of duties handled?	
17	Data protection	Provides appropriate encryption, retention and data handling controls.	Where is data processed and stored, and how is it protected?	
18	Pilot methodology	Provides a structured PoC using representative production documents.	What sample size, acceptance criteria and success measures do you recommend?	
19	Production readiness	Provides a credible path from pilot to governed production deployment.	What ongoing support, monitoring and change-management model is included?	

Scoring: 1 = inadequate, 3 = acceptable with gaps, 5 = strong fit. Weight criteria that are critical to your quality, traceability and integration requirements.

15. Pilot / PoC Checklist

A pilot should test real-world variability rather than a curated set of easy documents. The best pilot dataset includes clean digital PDFs, scans, different supplier layouts, multi-page certificates, tables, handwritten or stamped elements where relevant, missing fields and known exceptions.

- Use a representative document sample, including multiple suppliers and difficult layouts.
- Define baseline processing time and current exception/error categories.
- Agree field-level extraction and validation acceptance criteria before the pilot.
- Test known edge cases rather than only clean certificates.
- Test traceability from certificate to heat/lot/material/order.
- Test integration into the intended system of record.
- Measure exception volume and human review effort.
- Document security, access, retention and audit requirements.
- Define the production-readiness criteria before declaring the PoC successful.

16. What a Future-Ready MTR/COA Process Looks Like

A future-ready process does not simply replace manual data entry. It creates a controlled flow from incoming document to trusted operational data.

About Star Software

Star Software develops document automation solutions using capabilities across OCR, AI, RPA and deep learning. Its document automation use cases include MTR/COA processing, accounts payable automation, transcript processing and insurance document workflows.

The opportunity is to connect document intelligence with the systems and workflows that businesses already depend on—so documents become structured, usable and actionable information rather than another manual bottleneck.